

Generative AI Course - AWS

Notes

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Trends in AWS Cloud Computing

- Amazon Bedrock : is a fully managed service that offers a choice of high-performing foundation models (FMs) along with a broad set of capabilities that you need to build generative AI applications.
- Amazon Bedrock simplifies development with security, privacy, and responsible AI.

Trends in AWS Cloud Computing

- Amazon Q is a generative AI-powered assistant for work that is designed to be secure and private by default. Amazon Q is integrated into many AWS products.
- Two of the primary Amazon Q products are the following:
 - 1. Amazon Q Business : For Business Usage
 - 2. Amazon Q Developer : For Development

What is the Goal of Responsible AI ?

- Accurate, Fair, Safe and Secure
- Explainable and Transparent
- Governed and Controlled

Types of ML

- 1. Supervised : Labeled Data
 - 2. Unsupervised : Unlabeled Data
 - 3. Reinforcement : Rewards Based.
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- inference : An Inference is the Predictive output a model produces by applying what it has learned to input data.

Difference Between DL and Genrative AI

- Deep Leaning is type of ML That uses NN.
- DL models excel at processing large amounts of unstructured data to find complex relationships and respond within a domain.
- Generative AI is a type of Deep Leaning which is using DL model referred to as a Foundation Model.
- Foundation Models Train on Vast amounts of unstructured data and are able to provide generalized outputs.

AWS AI Services.

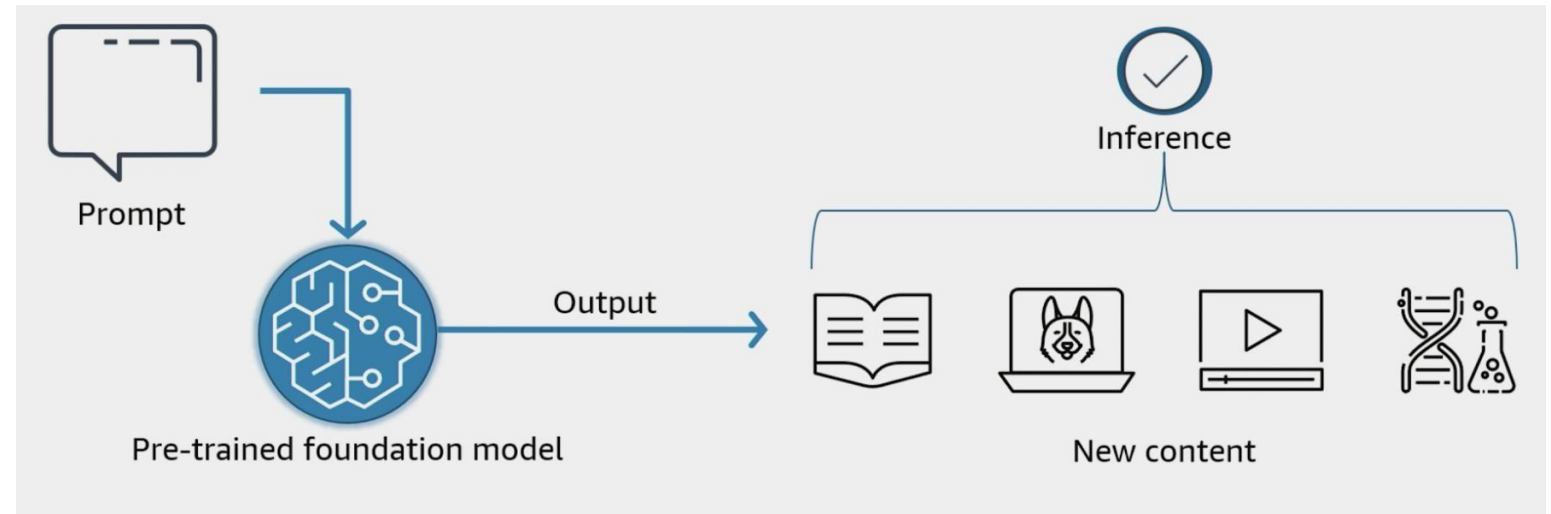
- Amazon Elastic Compute Cloud EC2 Trn1 & Inf1
- Amazon SageMaker
- Amazon Q
 - Amazon Q Business
 - Amazon Q Develop
- Amazon Bedrock
- Amazon Polly
- Amazon Transcribe
- Amazon Comprehend
- Amazon Translate
- Amazon Rekognition etc....

Demo : Using AWS AI Services

- Imagine One Situation : in your web application you have to provide Translation Service :
 - then Three AWS Services Can Combine and Used :
 - Amazon Transcribe : upload audio file. (From Amazon S3)
 - Amazon Translate : for Translation by NN.
 - Amazon Polly : to Convert Translated Text into Audio Again.

Foundation Models

- Common Categories of FM and GenAI
 - Language
 - Image
 - Audio
 - Biomedical



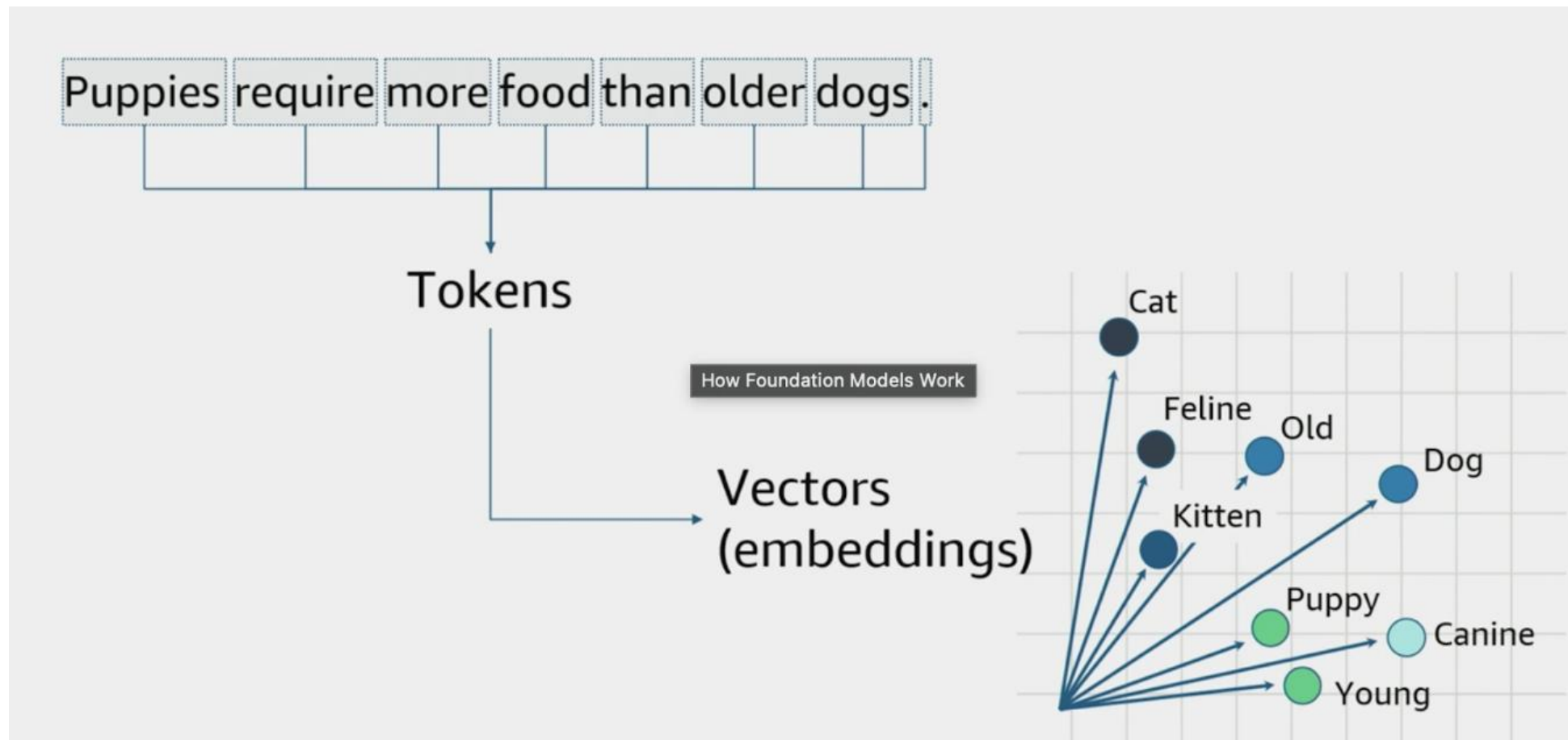
LLM are FMs for Text

- Prompt + LLM
 - Can do :
 - Text Generation
 - Question & Answer
 - Summation
 - Text Extraction
 - Conversations
 - Code Generation

Working of FMs

- Common Approaches to building Generative AI models :
 - Diffusion Models : The Models learn to Generate new Data by modeling the process of adding noise to a clean input then learning to reverse that process.
 - GANs : (Generative Adversarial Networks) : GANs Consist of two NN Trained Competitively.
 - Analogy : One tries to fool the other.
 - VAEs : Variational autoencoders : VAEs learn a latent representation of input data and generate new samples that resemble the original.
 - Transformer Based Models : These Models can focus on the most relevant parts of an input and pay attention to different parts of input simultaneously.

Tokenization

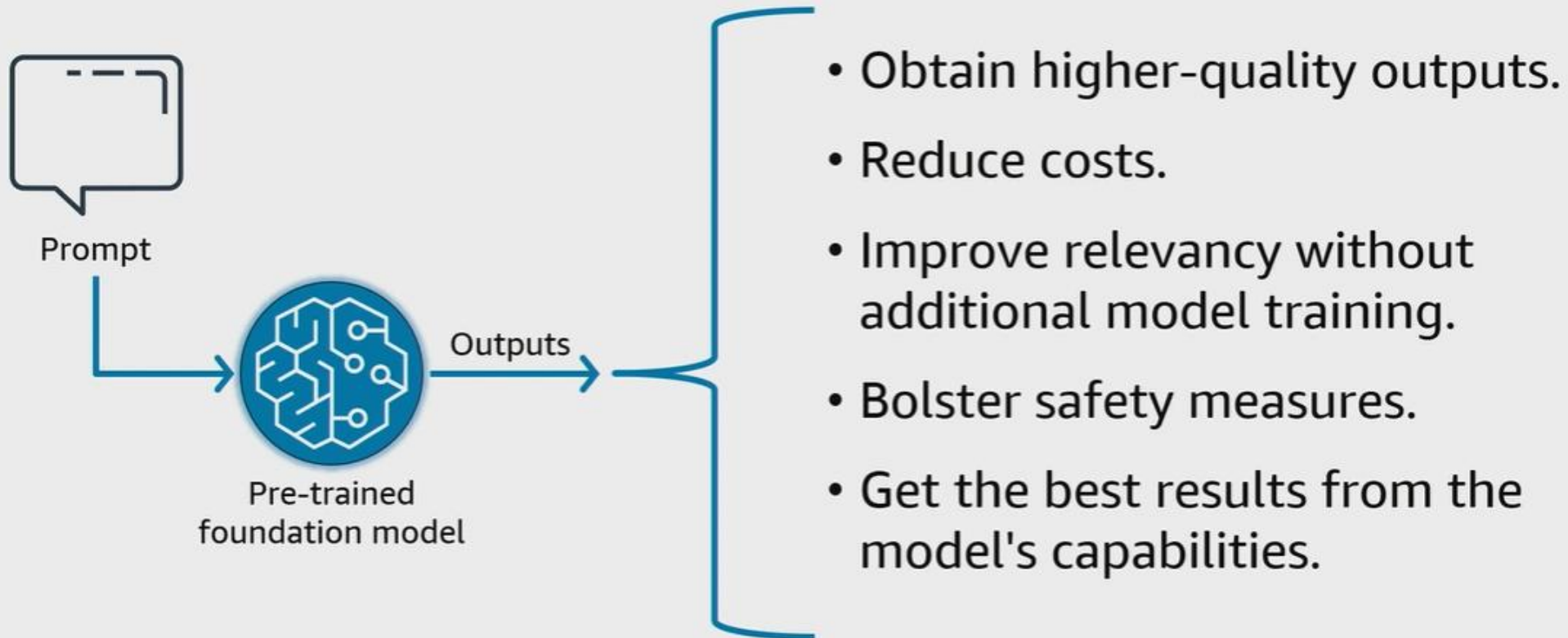


The Challenges of using outputs generated from FMs

- 1. Hallucinations
- 2. Toxicity
- 3. Bias
- 4. Illegality
 - if u want better output from Generative AI
 - Then u can use this :
 - Provide Information in this Prompt + Reference another data source + Perform Additional Training by using a task specific dataset.
 - Key Componentfor getting optimal O/P.
 - Model Choice
 - Prompt Design
 - Data Quality
 - Safeguards

Prompt Eng.

Well-designed prompts lead to better outputs



Common Elements to include in Prompts

- Task Description
- Instruction or Guidance
- Step by Step Instructions
- Response Format or Output Characteristics.
- **Customize The Nature of the Response :**
 - Context or Scenario
 - Input Data
 - Examples
 - Persona or Role
 - Audience
 - Tone and Style
 - Constraints or Exclusions

Prompting Techniques

- Zero Shot
 - Single Shot
 - Few Shot
 - Tool Based Prompting
 - Action Oriented Prompting
 - Negative Prompting
 - Chain of Thought
 - Output Refinement Techniques
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- Detailed Explanation : <https://www.promptingguide.ai/techniques>

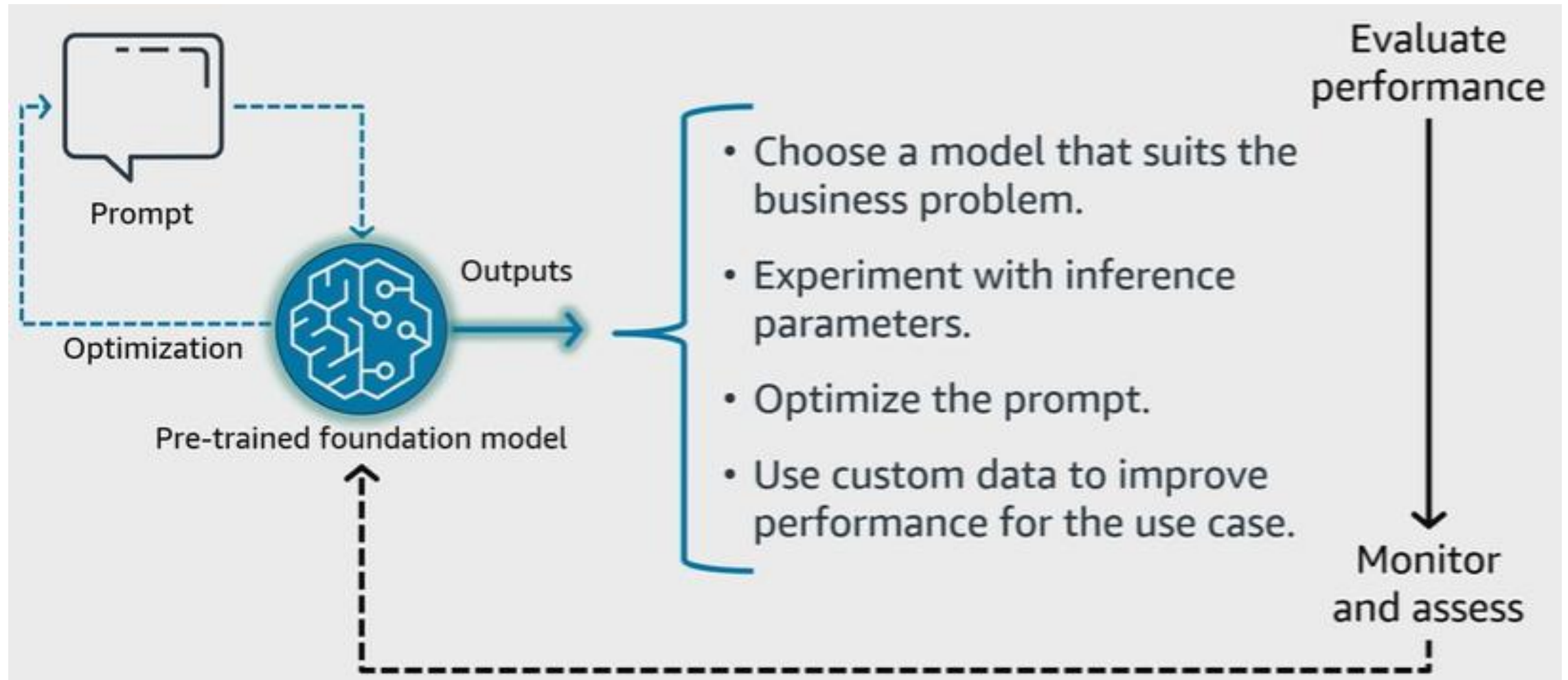
The Risk of Adverse Prompts

- Adverse Prompts : Malicious or unintended inputs that Attempt to manipulate the behaviour of Generative AI Systems.
- Which Requires Robust Security and Mitigation.
- Common Types of Adverse Prompts :
 - Prompt Injection
 - Prompt Leaking
 - Jailbreaking
 - Social Engineering

Foundation Model Approaches for our Generative AI Application

- Reuse (off-the Shelf)
- Adapt with Data
- Build From Scratch

Getting the most out of pre-trained FMs



Evaluating Models

- Define Metrics
- Prepare Test Datasets
- Test and Compare
- Iterate and Refine

Example Evaluation metrics for LLMs

- Perplexity : measures how well the model can predict the next word in a sequence of text.
 - Lower Scores indicate better performance.
- ROUGE : Recall-Oriented Understudy for Gisting Evaluation
 - measures how well an LLM generates text compared to a reference text.
 - this metric focuses on exact matches between texts and is primarily used for summarization tasks.
- BERTScore : measures how well language model's output matches reference text.

Example Evaluation metrics for LLMs

- BLEU : Bilingual Evaluation Understudy
 - Compares a generated sentence to one or more reference sentences.
 - This metric is used primarily to evaluate translation.
- Semantic Robustness : measures how much your model output changes as the result of small, semantic - preserving changes in the input.

Ongoing Monitoring and Assessment

- Monitor Performance in Production env.
- Collect and Analyze User's Feedback.
- Stay informed about new model releases.
- Periodically reassess.
- Consider Multi-Model Approaches.

Configuring Inference Parameters

- Inference Parameters is taking care of - “HOW MODEL IS LOSSLY AND TIGHTLY PREDICTING THE OUTPUT!”
- Example : The Cat Jumped onto the.....
 - in this example inference param will decide either an answer will more creative or more deterministic.
- so, Three Parameter will Impact more...

Adjusting randomness and diversity

Temperature

Controls the randomness of the output.

A higher value results in more diverse and creative outputs.

A lower value leads to more deterministic and focused outputs.

Range 0–1

Top P

Controls the cumulative probability of tokens to include as top candidates for the next token. P is a percentage of tokens.

A higher value includes more tokens, leading to more diverse outputs.

A lower value focuses on the most probable tokens, resulting in more coherent outputs.

Range 0–1

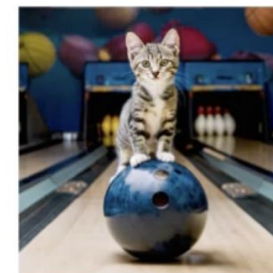
Top K

Restricts the sampling to the top K most probable tokens for the next token. K is a specific number of tokens.

A higher value includes more tokens in the sampling pool, leading to more diverse outputs.

A lower value focuses on only the top K tokens, resulting in more deterministic outputs.

Range 1–K



Higher =
more
creative



Lower =
more
deterministic

Configuring Inference Parameters

Setting Limits

- Max Length : is the Max Number of tokens that a model can generate in a single response.
 - This Parameter Controls the length of the output text produced by the model.
- Stop Sequence : are specific tokens or strings that signal the model to stop generating further tokens.
 - this parameter is used to control when the model should halt it's output generation.

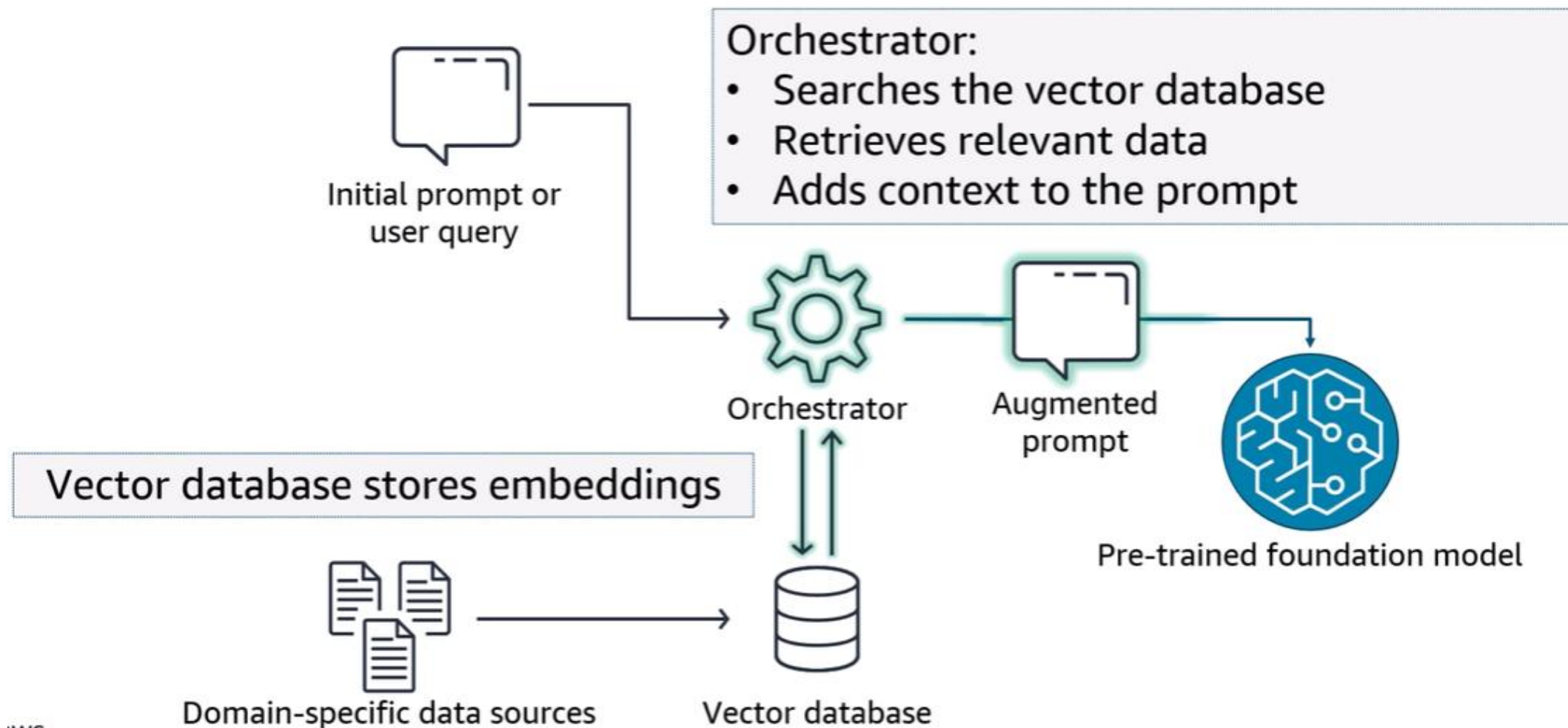
Introducing RAG : Retrieval-Augmented Generation

- The Process of Referencing an authoritative knowledge base outside an LLM before generating a response for a prompt.
- Combines the precision of RAG Systems with the power of an LLM.
- Augments the Prompt with Additional Context.
- Key Components of RAG :
 - Orchestrator
 - Vector Database : Stores Domain Specific Data Sources,

BASIC FLOW OF RAG

- Initial Prompt or User Query will be Sent to an Orchestrator where an Orchestrator will be Connected with the Vector Database.
- Now Vector Database is Connected with the Domain Specific Data Sources.
- Taking the Knowledge of Vector Database an Orchestrator (which is also connected with the pre-trained foundation models) will perform or do these operations :
 - Searches the vector Database
 - Retrieves Relevant Data
 - Adds Context to the Prompt

Diagram by AWS Academy



Amazon Bedrock Knowledge Bases

Implement RAG without building custom integrations

- Provides a fully managed RAG workflow
- Retrieves data from structured or unstructured sources
- Stores unstructured sources into a vector embedding
- Provides citations for responses
- Supports choices of parsing options, embedding models, and vector stores
- Provides a “chat with your document” feature to get responses without setting up a vector store



Amazon Q Business

Build an interactive chat application that references your data without coding

- Is powered by Amazon Bedrock
- Takes advantage of the controls implemented in Amazon Bedrock to enforce safety, security, and the responsible use of AI
- Uses your enterprise data or a combination of your data and LLM knowledge as the data source
- Supports a variety of LLM tasks
- Provides for integrated access control
- Generates source citations with each answer



Pre-trained FM Customization Techniques

- Fine Tuning
- Instruction Based Continued Pre-training
- Let's See Comparision table given by [CHATGPT.](#)

<u>Aspect</u>	<u>Fine-Tuning</u>	<u>Instruction-Based Continued Pre-training (ICPT)</u>
Definition	A process of training a pre-trained model on a specific labeled dataset to perform a particular task.	A process of continuing the pre-training of a language model using large collections of instruction-response data before task-specific fine-tuning.
Purpose	To adapt the model for a specific task such as classification, translation, or question answering.	To improve the model's ability to understand and follow natural language instructions across many tasks.
Training Data	Task-specific labeled datasets.	Large-scale instruction-following datasets consisting of prompts and responses.
Learning Objective	Minimize task-specific loss for better performance on a target task.	Learn instruction-following behavior while retaining general language understanding.
Scope	Narrow and focused on one or a few tasks.	Broad and general, improving performance across multiple instruction-based tasks.
Data Size	Usually small to medium-sized datasets.	Usually very large datasets with millions of instruction-response pairs.
Flexibility	Optimized for a specific application.	More flexible and capable of handling unseen instructions.
Generalization	Limited to tasks similar to the training data.	Better generalization to new and diverse tasks.
Training Cost	Lower computational cost.	Higher computational cost due to extensive continued pre-training.
Model Behavior	Improves accuracy on a particular task.	Improves the model's ability to follow user instructions naturally.
Examples	Sentiment analysis, spam detection, medical diagnosis, image classification.	Models such as InstructGPT, FLAN, T0, and instruction-tuned LLaMA models.
Advantages	High performance on specialized tasks, efficient for domain-specific applications.	Better instruction following, improved zero-shot and few-shot learning, enhanced general usability.
Limitations	May overfit to the target task and lose generalization ability.	Requires large computational resources and high-quality instruction datasets.

Defining Responsible AI

- Responsible AI Approach to building and using AI System That Encompasses the Following Principles :
 - People - Centric
 - Ethical
 - Safe
 - Unbiased
- Challenges to Responsible AI when using Generative AI :
 - Hallucinations
 - Toxicity
 - Bias
 - Illegality

Security, Governance & Compliance

- Security : Ensure the confidentiality, integrity and Availability of data, models and infrastructure.
- Governance : Establish the principles, policies and processes for managing AI Systems throughour their life cycle.
- Compliance : Ensure adherence to relevant regulations, industry standards and internal policies.

AWS Services for Security at Various Levels

- Compliance and Governance
 - AWS well Architected Framework Security Pillar
 - Generative AI Security Scoping Matrix
 - AWS Trusted Advisor
 - AWS WA Tool
- The Detection and Incident Response
 - Amazon CloudWatch
 - AWS CloudTrail
- Infrastructure Protection
 - Purpose-Built AI Infrastructure.

AWS Services for Security at Various Levels

- Identity and Access Management
 - AWS Identity and Access Management (IAM)
- Data Protection
 - AWS Key Management Service (KMS)
 - Amazon Simple Storage Solution (S3)
 - Amazon Macie

Amazon Inspector and AWS Artifact for compliance

Amazon Inspector

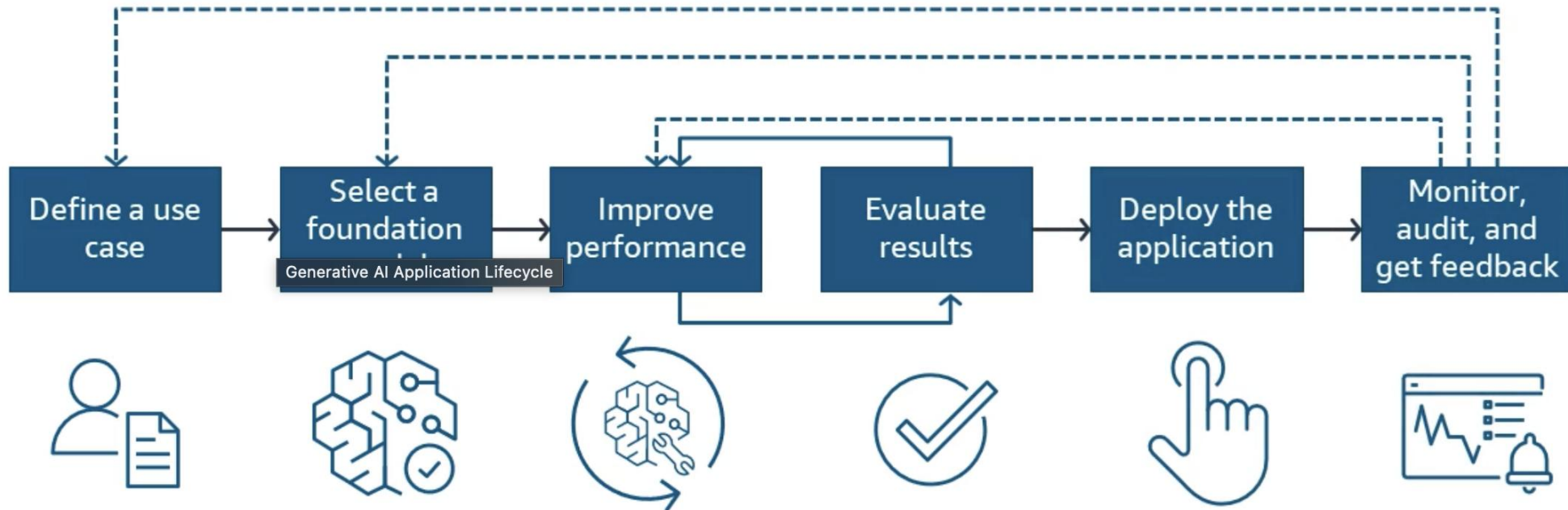
- Performs automated security assessments for AWS resources and applications
- Identifies vulnerabilities, exposures, and deviations from best practices
- Provides detailed, prioritized security findings
- Offers integration with development and deployment pipelines

AWS Artifact

- Provides on-demand downloads of AWS security and compliance documents
- Gives users the ability to download certifications from accreditation bodies that validate AWS security controls
- Gives users the ability to download security and compliance documents for independent software vendors (ISVs) on AWS Marketplace
- Supports submitting AWS Artifact documents to auditors or regulators as audit artifacts to demonstrate security and compliance

Generative AI Application Lifecycle

Generative AI application lifecycle



References

- 1. AWS Academy Generative AI Course.
- 2. Multiple AI LLM's to Understand to Concepts more deeply.
- 3. [promptingguide](#)

Thank you

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